

ViTon-GUN: Person-to-Person Virtual Try-on via Garment Unwrapping

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I. INTRODUCTION

Abstract—The image-based Person-to-Person (P2P) virtual try-on, involving the direct transfer of garments from one person to another, is one of the most promising applications of human-centric image generation. However, existing approaches struggle to accurately learn the clothing deformation when directly warping the garment from the source pose onto the target pose. To address this, we propose Person-to-Person virtual try-on via Garment UNwrapping, a novel framework dubbed as ViTon-GUN. Specifically, we divide the P2P task into two subtasks: Person-to-Garment (P2G) and Garment-to-Person (G2P). The P2G aims to unwrap the target garment from a source pose to a canonical representation based on *A-Pose*. In the P2G stage, we enable the implementation of a flow-based P2G scheme by introducing an *A-Pose* estimator and establishing comprehensive training conditions. Building upon this step-wise strategy, we introduce a novel pipeline for P2P try-on. Once trained, the P2G strategy can serve as a “plug-and-play” module, which efficiently adapts existing diffusion-based pre-trained G2P models to P2P try-on without further training. Quantitative and qualitative experiments demonstrate that our ViTon-GUN performs remarkably well on P2P try-on, even for dresses with intricate design details.

Index Terms—Person-to-person try-on, garment-to-person try-on, garment unwrapping, *A-pose* estimator, plug-and-play.

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VIRTUAL try-on aims to reasonably place a given garment image onto a person with a specific pose. In recent years, there have been plenty of works [1], [2], [3], [4], [5], [6] that have elaborately designed models that use in-shop garment images as references to achieve this goal. These works effectively map in-shop garments onto human images, a process termed Garment-to-Person (G2P) in this study. It is worth noting that in such G2P methods, the garment images used, referred to as in-shop garments, are similar to those displayed on store shelves. They are flattened and have a standardized pose, which we term as the *A-Pose* in this study. The *A-Pose* enables the complete display of the front of the garment in the images. However, in practical application scenarios, the direct transfer of clothing from one arbitrary pose to another, a process known as Person-to-Person (P2P) try-on, exhibits a wider and more urgent business demand in comparison to G2P try-on [7]. Unfortunately, research is scarce on P2P try-on [8], [9]. The main challenges [10] stem from the adverse effects resulting from arbitrary poses in garment images, including issues such as dragging, occlusion, distortion, and tilting, which greatly impede the seamless and precise transfer of the target clothing to the desired pose. Another key problem [11] is the difficulty in obtaining a dedicated dataset for P2P try-on. Such a dataset requires each sample to consist of more than two poses of the same person wearing the same garment.

To address the problem led by the pose changing, Cui [12] and Huang [13] propose flow-based models to directly warp the garment from the source pose to the target pose. Their motivation explicitly utilizes more conditional information to model the complex and diverse nonrigid deformation of garments from various poses. While effective for handling poses without significant transformation, their methods struggle to maintain high fidelity of the try-on results and cannot accurately model clothing deformation depending on the degree of transformation. Additionally, Xie [7], [14] proposed a method to mitigate the influence of the source pose on the garments by simply segmenting the clothes into patches and then reassembling them onto the target pose. However, this segmentation step is manual and non-learnable. When transferring these clothes patches to another model, aligning and stitching them seamlessly, precisely, and realistically becomes challenging. To improve the try-on fidelity, recent research works leverage the remarkable generative capabilities of diffusion models [10], [11], [15], [16]. These works currently train their models on high-quality and standard virtual try-on benchmarks specific to G2P tasks, such as



Fig. 1. Exemplary results of ViTon-GUN. We present a novel perspective for Person-to-Person (P2P) try-on by decomposing it into Person-to Garment (P2G) and Garment-to-Person (G2P) subtasks.

VITON-HD and Dress Code. In addition, TryOnDiffusion [17] has collected a dataset containing 4 Million pose pairs specific to P2P task, it has not been made available to the public yet.

In this paper, we aim to investigate a pipeline that effectively adapts the existing high-quality and published G2P try-on datasets to P2P try-on. In practice, as illustrated in Fig. 1, we tackle the Person-to-Person (P2P) problem by decomposing it into two steps: Person-to-Garment (P2G) and Garment-to-Person (G2P), effectively addressing the aforementioned challenges simultaneously. In the P2G step, the garments from the source pose are unwrapped by fitting them to the *A-Pose*, which is free from any obstructions or distortions. In this way, the unwrapped garments in Fig. 1 are aligned to in-shop garments in G2P benchmarks. By doing so, due to eliminating the adverse effects introduced by the source pose, the P2P task is expected to become more explicit and easier to implement. More importantly, we align the P2P tasks with G2P tasks, thereby utilizing open-source and high-quality databases specific to G2P tasks to address P2P tasks effectively.

To the best of our knowledge, there is currently no relevant research work that can transform clothes from a source pose to an *A-Pose*. Motivated by the warping process [3], [6], [18] used in G2P, we define the unwrapping of the garment from the source pose back to a flat state under the *A-Pose* in the P2G as the inverse warp in the G2P. However, achieving this reverse operation presents considerable challenges. Specifically, most of the existing G2P approaches employ appearance flow estimation [19], [20] into neural networks. However, the flow-based methods require the use of target pose information, such as key points or dense pose, as guiding signals to warp the target garment to the target pose. Therefore, without the target information of the *A-Pose* as a guiding signal, we cannot implement the inverse warp to unwrap the source garment in the P2G task. To address this issue, we design an *A-Pose* estimator to predict the corresponding key points for each garment, thus effectively implementing the reverse operation of G2P.

Furthermore, by introducing the *A-Pose* as a canonical representation in the P2P task, rather than directly warping the garment from the source pose to the target pose, our pipeline is capable of handling large pose changes and reducing the

dependency on pose changing while preserving garment details and shape. It should be noted that we utilize publicly available and easily accessible datasets [5], [21] that contain in-shop garments (also referred to as *A-Pose* garments here) to train our P2G method, thus avoiding the critical issue of lacking dedicated datasets. The P2G strategy thoroughly eliminates the negative impact from the source poses and normalizes the cloth from a random source pose to a canonical representation. Additionally, we warp the unwrapped garment into the target pose by using a cascaded clothing flow estimation model. Finally, we propose a Style-GAN-based generator with dual-path shape-texture embedding to generate high-quality images.

Our contributions can be summarized as follows.

- We present a novel perspective for addressing the P2P task by decomposing it into P2G and G2P steps, relying solely on publicly available and easily accessible datasets. This makes the challenging P2P task more tractable and achievable.
- We propose an efficient P2G scheme that can unwrap target clothes to an *A-Pose* garment while preserving fine details and ensuring that their original shape is intact, completely removing the dragging, distortion, and occlusion from the source pose.
- We introduce a novel pipeline consisting of the P2G, Warp, and generation modules with comprehensive data optimization processing techniques.
- Our P2G strategy can also function as a plug-and-play adapter enabling numerous state-of-the-art diffusion-based methods designed for G2P tasks to be directly applied to P2P tasks, achieving the outperformance with only the corresponding pre-trained models.

II. RELATED WORKS

Fashion Generation: Fashion generation has attracted increasing attention recently thanks to the rapid advancement of AI-based content generation. The existing methods [22], [23], [24], [25], [26], [27] have demonstrated that deep learning has tremendous potential in the fashion generation field. ARMANI [28] proposes a part-level-based garment-text alignment method which allows bi-directional conversion between

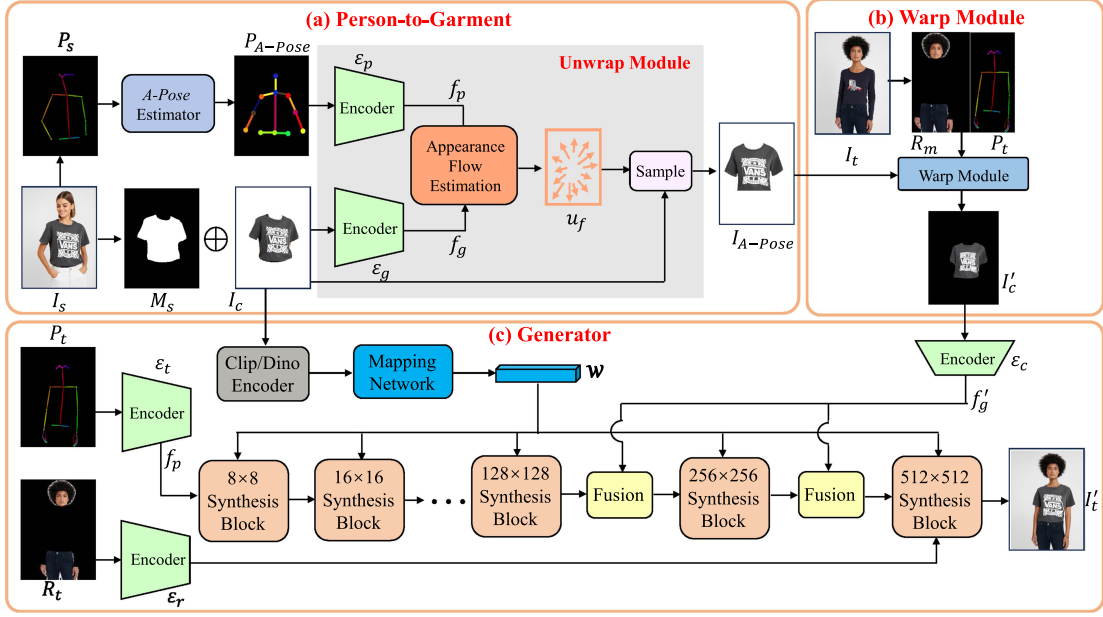


Fig. 2. Overview of ViTON-GUN. The Person-to-Garment stage aims to unwrap the garment I_c from source person I_s to canonical representation with A -Pose I_{A-Pose} . In this stage, the introduction of the A -Pose estimator enables the application of Appearance Flow Estimation to this task by establishing comprehensive training conditions. The Warp module warps the clothing I_{A-Pose} to the target pose P_t and sends the warped garment I'_c containing shape information and boundary alignment to the generator. The generator embeds the texture and shape information into the synthesis block through a dual-path, resulting in the generation of the try-on image.

garment images and text descriptions. SGDiff [29] introduces a style-guided diffusion model for text-to-image fashion synthesis. DiffCloth [30] provides the diffusion model with flexible combinability in the fashion domain by structurally adapting the cross-modal semantics. DiffFashion [24] designs an unsupervised structure-aware transfer method utilizing the conditional diffusion model [31], [32] for semantically generating new garments from a given garment image and a reference appearance image.

G2P Virtual Try-on: G2P is the mainstream of the existing methods [3], [5], [33], [34], [35], [36], [37] for virtual try-on. VITON [4] first contributes a two-stage network structure to deform the source garment and fuse it with the target body, but its accuracy and quality are not high. CP-VTON [38] employs a geometric matching module to learn the deformation of garments in different poses, increasing accuracy and flexibility. VTON-GAN [39] uses multi-scale adversarial loss to further improve the robustness of G2P virtual try-on. Recent focus has shifted to diffusion approaches for their superior generation capabilities [17], [40], [41], [42]. LaDI-VTON [41] injects features of the source garment and caption into the pre-trained diffusion model [31], [32] to guide the generation. DCI-VTON [40] combines a warp module with an inpainting diffusion [31] model to accomplish garment deformation and interaction. Indeed, training these models is resource-intensive, however, the numerous pre-trained models released for G2P tasks mentioned above cannot be directly used to address P2P tasks. Achieving this would significantly advance P2P research and applications.

P2P Virtual Try-on: Compared to G2P virtual try-on, P2P virtual try-on is more practical in realistic scenes, since the in-shop garment similar to the data in VITON-HD is not always easily accessible in most cases. Nevertheless, the researches [8], [9], [12], [14] to tackle the P2P virtual try-on

is sorely lacking. Dior [12] adopts a recurrent neural network structure [43] to generate the parts of the garment in a certain order. Pose-with-Style [9] achieves the transfer and control from source image to target pose by texture-source correspondence and the modulation of StyleGAN [44]. PASTA-GAN++ [14] introduces a chunk-based garment representation and a chunk-guided parsing synthesis module, enabling the ability to handle arbitrary garment categories localized editing. Although Wear-Any-Way [11], TPD [10], CAT-DM [15] and StableVITON [16] claim to be applicable to P2P tasks, they require additional large-scale databases containing pose pair representations and retraining, rather than directly leveraging the currently released pre-trained models designed for G2P tasks.

III. METHOD

Given a source image I_s of a person wearing a garment I_c , and a target person image I_t , the P2P try-on aims to generate the image I'_t which retains the identity condition R_t obtained from I_t while wearing the source garment I_c . To achieve this, we decompose the task into three stages: unwrapping garment I_c from the source pose P_s to a canonical representation I_{A-Pose} , warping the garment I_{A-Pose} to target pose P_t and synthesize the final try-on image I'_t , as shown in Fig. 2. This strategy, introduced for the first time, unwraps the clothing I_c from source pose P_s to an A -Pose P_{A-Pose} , which essentially involves the inverse warp process in G2P Try-on (see Section III-A). To ensure the unwrapped cloth I_{A-Pose} reasonably matches the human structure, A -Pose keypoints are essential as a reference. Therefore, we propose a training-free A -Pose estimator to predict a set of A -Pose keypoints P_{A-Pose} corresponding to each keypoint from P_s . This creates a complete training setting that enables the inverse warp process to be achieved using a flow-based warp model. Moreover, we

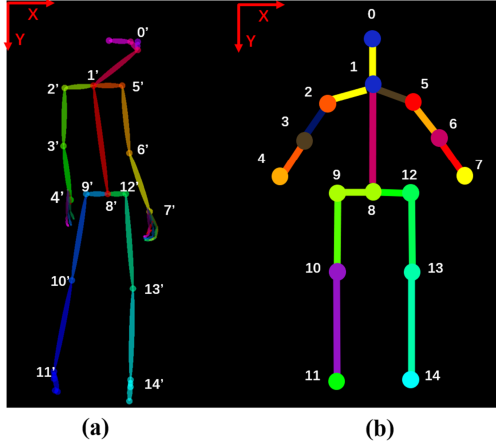


Fig. 3. A-Pose Estimator. (a) Keypoints of source pose, (b) Keypoints of A-Pose.

used a cascade appearance flow warping module to warp the A-Pose clothing I_{A-Pose} to target pose P_t and propose a StyleGAN-based generator with dual-path shape-texture embedding to synthesize the try-on image I'_t , which is introduced in detail in Sections III-B and III-C, respectively.

A. Person-to-Garment

Inspired by flow-based warp models for G2P try-on [6], [18], [20], which utilize the target pose features as guidance to warp in-shop garments to target poses, we can apply a similar strategy to warp garments from the source poses to fit A-Pose. However, in practice, the A-Pose does not have well-annotated keypoints that can be used as guidance in the warp process. Therefore, we propose an A-Pose estimator to calculate the specific 25 standard keypoints for A-Pose. Furthermore, we use the appearance flow estimation to warp the target clothing I_c from the person model I_s to be A-Pose garment I_{A-Pose} as A-Pose keypoints P_{A-Pose} guidance.

A-Pose Estimator: To apply the flow-based warp network to the P2G method, it is necessary to construct the A-Pose keypoints for each sample. Building upon the 25 keypoints map and their annotation order as suggested by Openpose [45], we extract the 14 essential feature points that have significant impacts on the shape of the clothes, as depicted in Fig. 3. These points include *Neck*, *MidHip*, *Shoulder*, *Hip*, etc., aiding in the unwrapping the garment from the source pose to the A-Pose. We have labeled the X and Y coordinate directions in Fig. 3 and conduct the A-Pose estimator to obtain A-Pose keypoints based on the following principles:

- 1) For *Neck* (point 1 in Fig. 3): The coordinates of *Neck* between two poses are the same: $(x_1, y_1) = (x_{1'}, y_{1'})$
- 2) For *MidHip* (point 8 in Fig. 3): The coordinate of *MidHip* in A-Pose could be obtained by: $(x_8, y_8) = (x_1, y_{8'})$
- 3) For *Elbow*, *Wrist* and *Shoulder* (point 3, point 4 and point 2 in Fig. 3): The angle and distance of the left part of these segments can be specified by:

$$\langle \vec{2\ 3}, \vec{3\ 4} \rangle = 0^\circ, \langle X, \vec{2\ 1} \rangle = 10^\circ \quad (1a)$$

$$\langle X, \vec{4\ 2} \rangle = 60^\circ \quad (1b)$$

$$|\vec{1\ 2}| = \max \left\{ |\vec{1'\ 2'}|, |\vec{1'\ 5'}| \right\} \quad (1c)$$

$$|\vec{2\ 3}| = \max \left\{ |\vec{2'\ 3'}|, |\vec{2'\ 6'}| \right\} \quad (1d)$$

$$|\vec{3\ 4}| = \max \left\{ |\vec{3'\ 4'}|, |\vec{6'\ 7'}| \right\} \quad (1e)$$

Here, X represents the horizontal direction, and Y represents the vertical direction, as shown in Fig. 3.

The right part of them should be the same.

- 4) For *Hip*, *Knee* and *Ankle* (point 9, point 10 and point 11 in Fig. 3): The angle and distance of left part of these segments can be specified by:

$$\langle \vec{9\ 8}, \vec{8\ 12} \rangle = \langle \vec{9\ 10}, \vec{10\ 11} \rangle = 0^\circ \quad (2a)$$

$$\langle \vec{9\ 8}, \vec{9\ 10} \rangle = 90^\circ \quad (2b)$$

$$|\vec{8\ 9}| = \max \left\{ |\vec{8'\ 9'}|, |\vec{9'\ 12'}| \right\} \quad (2c)$$

$$|\vec{9\ 10}| = \max \left\{ |\vec{9'\ 10'}|, |\vec{12'\ 13'}| \right\} \quad (2d)$$

$$|\vec{10\ 11}| = \max \left\{ |\vec{10'\ 11'}|, |\vec{13'\ 14'}| \right\} \quad (2e)$$

The right part of them should be the same.

Unwrapping Module: As shown in Fig. 2, our unwrapping module is composed of the pyramid feature extraction and the cascade appearance flow estimation. The clothing I_c is fed into a feature pyramid network ε_g to generate multi-scale garment feature while A-Pose information P_{A-Pose} is input to the encoder ε_p . The encoders ε_g and ε_p share the same architecture.

The cascade appearance flow estimation aims to predict the dense correspondences between the clothes image I_c and the A-Pose keypoints P_{A-Pose} for unwrapping the garment. The output of the appearance flow estimation is a set of 2D coordinate vectors u_f . Each vector indicates which pixels in the clothes image I_c should be used to fill the given pixel in the A-Pose Garment I_{A-Pose} . Here, we adopt the style-based appearance flow estimation module [18] to estimate the flow vectors u_f .

The predicted appearance flow u_f from the style-based appearance flow estimation module is used to unwrap the garment:

$$I_{A-Pose} = S(I_c, u_f) \quad (3)$$

where $S(\cdot, \cdot)$ is the sampling operator.

To train the unwrapping module, the perceptual loss [46] between I_{A-Pose} and the in-shop ground truth clothing I_{gt} is adopted:

$$L_p = \sum_i \|\phi_i(I_{A-Pose}) - \phi_i(I_{gt})\|, \quad (4)$$

where ϕ_i is the i_{th} block of the pre-trained VGG network [47].

Following the standard in previous flow estimation approaches [6], [18], [20], the smoothness regularization on the predicted flow vectors and the loss on unwrapped garment I_{A-Pose} is also used as follows:

$$L_{smooth} = \sum_i \|\nabla_i(u_f)\| \quad (5)$$

$$L_M = \|I_{A-Pose} - I_g\| \quad (6)$$

The overall loss functions for the person-to-garment stage can be referred to as follows:

$$L_{overall} = \lambda_p \cdot L_p + \lambda_s \cdot L_{smooth} + \lambda_m \cdot L_M, \quad (7)$$

where λ_p , λ_s , and λ_m are hyper-parameters for balancing the three terms.



Fig. 4. Shoulder-to-torso alignment. (a) Stretched garment, (b) Source person, (c) In-shop garment, (d) Aligned garment.

Shoulder-Torso Alignment: In the P2G stage, if we directly use the in-shop garment (see Fig. 4(c)) as the supervisory signal to unwrap and restore the garment from the source person, it will cause serious texture stretching as shown in Fig. 4(a). From Fig. 4(b) and (c), The visible part of the garment on the person is smaller than what is presented in the in-shop garment because the area marked by black dashed lines in the in-shop garment correspond to the person's body but is not visible.

Flow-based methods [2], [3], [18], [48] warp the input garment according to the preserved region mask and force the warped garment to align with the boundary of the preserved region. In this situation, the appearance flow estimation model in the unwrapping module will stretch the texture to align with the larger shape boundary of the in-shop garment. This means that the boundaries marked by orange lines in Fig. 4(b) are forced to match the boundaries marked by orange lines in Fig. 4(c). Therefore, aligning the in-shop garment to the visual size of the clothing on the source person is crucial for preserving texture in the person-to-garment stage. To tackle this problem, we have developed a Shoulder-Torso alignment criterion by defining the Shoulder-to-Torso ratio, which is represented as $\frac{L_s}{L_t}$. The torso portion of the in-shop garment will be trimmed from the bottom to ensure it has the same Shoulder-to-Torso ratio as the garment on the source person.

B. Warping Module

The warping module and the unwrapping module have the identical architecture except for the difference in the inputs. The warping model takes as inputs the augmented garment I'_{A-Pose} and target retain image cascaded with target pose P_t . In the training phase, the garment parse predicted by an off-the-shelf human parsing model is used as the supervisory signal.

To train the warping module, the perceptual loss [46] between the output I'_c of the warping module and the cloth parse I_c predicted by an off-the-shelf human parsing model is adopted:

$$L_{pw} = \sum_i \|\phi_i(I'_c) - \phi_i(I_c)\|, \quad (8)$$

where ϕ_i is the i_{th} block of the pre-trained VGG network [47].

In addition, we also adopt the smoothness regularization on flow vectors u'_f and restored loss on warped garment I'_c as following.

$$L_{smoothw} = \sum_i \|\nabla_i(u'_f)\| \quad (9)$$

$$L_{Mw} = \|I'_c - I_c\| \quad (10)$$

The overall loss function for the warping module can be represented as follows:

$$L_{overallw} = \lambda_{pw} \cdot L_{pw} + \lambda_{sw} \cdot L_{smoothw} + \lambda_{mw} \cdot L_{Mw}, \quad (11)$$

where λ_{pw} , λ_{sw} and λ_{mw} are hyper-parameters.

C. Generator

Inspired by the remarkable capabilities of StyleGAN2 [44] in image synthesis, we have developed a dual-path shape-texture embedding generator, which builds upon the core architecture of StyleGAN2 and extends it to a conditional version as shown in Fig. 2. Considering that the unwrapping garment step and warping garment step will inevitably result in some loss of texture information, in the generator, we introduce an image embedding of the original garment I_c extracted by a CLIP [49] image encoder to compensate for the texture information loss caused by these two steps. We define this branch as the CLIP path. In details, the garment I_c is projected onto ω through a pre-trained clip image encoder [49] followed by a mapping network, which helps in preserving more color and texture information from the original garment. It is worth noting that our design is not bound to a specific encoder (e.g., CLIP). Our model can be readily enhanced by incorporating more recent advanced image encoders, such as DINO-V2 [50]. The 8×8 synthesis block takes as input the pose features f_p and leverages the style code ω as the injected vector for synthesis blocks from 8×8 to 512×512 to generate the coarse try-on synthesis which guarantee the consistence of global color and texture between original garment and the synthesized result.

Through the warp network, the shape and texture of garment I'_c are aligned and transformed to the target pose. To generate the try-on image with accurate shape and texture details, the multi-scale features f'_g are injected into the synthesis blocks after the 128×128 by fusion models as shown in Fig. 2. The fusion models contains three SPADE Residual Blocks from [51]. In addition, to ensure identity information R_t (such as the head, hands, and feet) from the target person I_t to be fully transferred to the try-on result I'_t , we introduce an identity encoder ε_r to extract the multi-scale identity features and embed the information into the final synthesis blocks.

As paired images of models wearing garment I'_c simultaneously are not unavailable, our generator is trained on paired images of the same model, where one image depicts the model with the garment I'_c masked out, and the other image shows the model wearing I'_c . These paired images are used as training data. During this process, both reconstruction loss and perceptual loss [46] are used.

$$L_{recg} = \|I_t - I'_t\| \quad (12)$$

$$L_{pg} = \sum_i \|\phi_i(I_t) - \phi_i(I'_t)\|, \quad (13)$$

where ϕ_i is the i_{th} block of the pre-trained VGG network [47].

Additionally, we compute the adversarial loss L_{GANg} for image I'_t using the same approach as StyleGAN2 [44].

$$L_{overallg} = L_{GANg} + \lambda_{recg} \cdot L_{recg} + \lambda_{pg} \cdot L_{pg}, \quad (14)$$

where λ_{recg} and λ_{pg} are hyper-parameters for balancing the three terms.

IV. EXPERIMENTS

A. Datasets and Baselines

We train the unwrapping module, the warping module and the generator module separately and combine them to generate the P2P try-on image eventually. The in-shop data are only used in training the unwrapping module. In the testing process, we use the source person and target person images to evaluate the performance of our pipeline. Our experiments are conducted on three public and high-resolution G2P try-on benchmarks VITON-HD [21], Deepfashion [52] and DressCode [5] under the resolution of 512×384 . VITON-HD contains 13,679 image pairs of front-view upper-body women and in-shop upper garments. For Deepfashion, we utilized the unwrapping model pre-trained on VITON-HD to generate unwrapped garments for upper items. We exclusively consider upper items with front-view cases featuring full-body and half-body persons and randomly select 20% of these cases as test pairs, resulting in a total of 2346 test pairs. Of note, in the Deepfashion dataset, we did not utilize any in-shop garment data.

In order to examine the effectiveness of our model with dresses data, we also conducted experiments on dresses sub-dataset from DressCode Dataset. All pre-processing dresses data used in this study are from GP-VITON [2].

For the P2G step, we compare our unwrapping approach with VAE [53], which is trained from scratch on VITON-HD and the dresses sub-dataset from the DressCode Dataset. In addition, for the P2P task, we compare our pipeline with several state-of-the-art methods, including flow-based approaches (Dior [12] and FWS [9]), Style-GAN-based method PASTA-GAN++ [14], and diffusion-based approaches CAT-DM [15] and StableVITON [16]. Dior and FWS models are pre-trained on the Deepfashion dataset [54], while PASTA-GAN++ models are pre-trained on the UPT dataset [14]. Both DeepFashion and UPT contain upper-body garments and dress items. Additionally, the comparative analysis experiments with CAT-DM were conducted based on CAT-DM's pre-trained models on the VITON-HD and DressCode dataset. Due to StableVITON having pre-trained models only on the upper-body cloth and not trained on dress items, we only conducted comparative experiments with StableVITON on VITON-HD and did not perform comparative analysis with StableVITON on the dresses sub-dataset of DressCode.

B. Implementation Details

Unwrapping Module and Warping Module: The warping module and the unwrapping module use the same training hyper-parameters. In detail, both the unwrapping module and the warping module are implemented in PyTorch [55]. We set the batch size as 10 and trained both models with 200 epochs each using the Adam optimizer [56]. The initial learning rate is set to 1×10^{-5} which is decayed after 50 epochs. The hyper-parameters λ_p , λ_s and λ_m are set to 0.2, 5.0 and 1.0, respectively. The hyper-parameters λ_{pw} , λ_{sw} and λ_{mw} also are set as 0.2, 5.0 and 1.0, respectively.

Generator: When training the generator, we set the batch size as 16 and trained the model for 4 million iterations with a



Fig. 5. Qualitative comparison on upper-body for P2G on VITON-HD.

learning rate of 1×10^{-4} using the Adam optimizer [56]. Both the loss hyper-parameters λ_{recg} and λ_{pg} are set as 40. Upon the acceptance of this work, we will make the code available for release.

C. Quantitative Comparison on P2G

For P2G, we adopt FID [57], KID [58], SSIM [59] and LPIPS [60] to measure the accuracy and realism of the unwrapped garments. In addition, we invite 30 volunteers to complete the 30 tasks by choosing the most realistic unwrapped garment given personal images. Finally, the human evaluation (HE) score is calculated as the chosen percentage for each method. As reported in Table I, our method significantly outperforms VAE in terms of the FID, KID and HE. The experimental results demonstrate that the VAE is not an effective solution for the P2G task.

D. Qualitative Comparison on P2G

Figs. 5 and 6 provide a comprehensive visual comparison between our unwrapping approach and VAE for upper body clothing and dresses, respectively. Even when dealing with challenging poses, such as the second case in the first row of Fig. 5, where a lady is placing her hands on her hips, our method is still capable of restoring the complete shape and details of the garment for upper-body clothing. Furthermore, our method is capable of restoring texture details such as stripes, preserving fine-grained information. For dresses, our method can realistically restore rich details in cases such as spaghetti straps, one-shoulder designs, and skirt details. On the contrary, VAE falls significantly short in restoring the texture and shape information of the garment. To validate the generalization of our unwrapping model, we utilized the unwrapping model trained on VITON-HD to infer unwrapping garments in the Deepfashion dataset, as shown in Fig. 7. It is worth noting that in the Deepfashion dataset, we did not utilize any in-shop garments corresponding to the source poses.

TABLE I
QUANTITATIVE COMPARISON ON UPPER-BODY CLOTHING AND DRESSES FOR P2G ON VITON-HD, DRESSCODE AND DEEPPASHION

Method	VITON-HD					DressCode (Dresses)					Deepfashion
	FID ↓	KID ↓	SSIM ↑	LPIPS ↓	HE ↑	FID ↓	KID ↓	SSIM ↑	LPIPS ↓	HE ↑	HE ↑
VAE	49.61	2.921	0.706	0.184	0%	94.83	2.427	0.7737	0.3098	0%	0%
P2G(Ours)	20.15	0.881	0.792	0.152	100%	17.31	0.5142	0.7820	0.2250	100%	100%

For deepfashion, we used the pre-trained unwrapping model from VITON-HD to generate unwrapped garments. Note: in the deepfashion dataset, there is no in-shop garment data for the source poses, so quantitative metrics are not computable.



Fig. 6. Qualitative comparison on dresses for P2G on DressCode.

TABLE II
QUANTITATIVE COMPARISON ON UPPER-BODY CLOTHING AND DRESSES FOR P2P ON VITON-HD AND DRESSCODE

Method	VITON-HD			DressCode(Dresses)		
	FID ↓	KID ↓	HE ↑	FID ↓	KID ↓	HE ↑
Dior	27.81	2.484	2.278%	70.31	6.817	1.040%
PWS	15.70	0.618	5.684%	20.83	0.821	2.346%
PASTA-GAN++	10.94	0.248	17.80%	20.46	1.011	12.67%
CAT-DM	11.89	0.360	6.291%	25.98	1.230	8.425%
P2G+CAT-DM	11.59	0.338	8.714%	23.74	0.985	11.46%
StableViton	11.78	0.383	8.250%	-	-	-
P2G+StableViton	10.45	0.258	18.29%	-	-	-
ViTon-GUN (CLIP)	10.06	0.167	32.68%	11.67	0.146	64.04%
ViTon-GUN (DINO)	10.00	0.154	-	11.61	0.144	-

E. Quantitative Comparison on P2P Try-On

Tables II and III present the quantitative comparisons between our proposed ViTon-GUN and state-of-the-art methods, including approaches designed for P2P tasks such as Dior [12], PWS [9], and PASTA-GAN++ [14], as well as diffusion-based approaches that are trained for the G2P task, such as CAT-DM [15] and StableVITON [16]. For CAT-DM [15] and StableViton [16], we use the clothing parse obtained from the source person image as the in-shop garment to adapt these diffusion-based methods to P2P application. Additionally, we incorporated our unwrapping module (also called P2G) as a plug-and-play adapter into StableVITON and CAT-DM, referred to as P2G+StableVITON and P2G+CAT-DM, respectively. Specifically, we replaced the in-shop garment required in StableVITON and CAT-DM with the unwrapped garment.

TABLE III
QUANTITATIVE COMPARISON ON UPPER-BODY CLOTHING ON DEEPPASHION DATASET

Method	Deepfashion		
	FID ↓	KID ↓	HE ↑
Dior	18.01	1.364	10.27%
PWS	26.67	2.308	1.210%
PASTA-GAN++	16.57	0.464	11.64%
CAT-DM	14.28	0.429	11.57%
P2G+CAT-DM	13.65	0.361	11.83%
StableViton	13.38	0.401	13.20%
P2G+StableViton	12.56	0.297	13.06%
ViTon-GUN (DINO)	10.34	0.141	27.28%

For HE, we invited 30 volunteers to individually select the best realistic P2P try-on results from a set of 30 upper-body cases and 30 dress cases. As shown in Table II, directly applying state-of-the-art diffusion-based try-on methods to the P2P task has not yet achieved satisfactory performance levels. However, by integrating our P2G strategy into these diffusion-based methods as plug-and-play, their performance on the P2P task is significantly improved, especially for StableVITON, where the FID score has been significantly reduced from 11.78 to 10.45 on VITON-HD dataset. Note that for dress cases, our method demonstrates notable performance in terms of all evaluation metrics scores compared to the comparative methods. StableVITON is excluded from the comparisons on DressCode (dresses) as it is not trained on dress items.

F. Qualitative Comparison on P2P Try-On

Figs. 8 and 10 display the qualitative comparison of our ViTon-GUN with the state-of-the-art baselines on upper-body clothing [21] and dresses [5], respectively. The image results show that our method exhibits better performance in preserving the shape of the clothing and maintaining texture details in the synthesized try-on results, especially in preserving the original skirt shape of dresses, which stands out prominently.

Specifically, Dior and FWS cannot generate the clothing details and realistic try-on results effectively. By examining the results in the 6th row and 5th column of Fig. 8, as well as the sample in the 4th row and 5th column of Fig. 10, it is evident that PASTA-GAN++ encounters challenges in aligning and stitching garment patches seamlessly, precisely, and realistically when disentangling the source garment from the source pose. PASTA-GAN++ adopts a disentangling approach that involves dividing the source garment into patches, applying normalization transformations to these patches, and then stitching them onto the target pose. However, this disentangling strategy leads to inaccuracies in sleeve length and garment shape when generating try-on images. In contrast, our proposed method introduces the P2G step to disentangle the source garment from the source pose, addressing these issues. Diffusion-based models CAT-DM and StableVITON have exhibited promising capabilities G2P



Fig. 7. Qualitative comparison for P2G on Deepfashion dataset. We utilized the unwrapping model pre-trained on VITON-HD to generate the unwrapped garments, and we did not conduct unwrapping model training on the Deepfashion dataset.

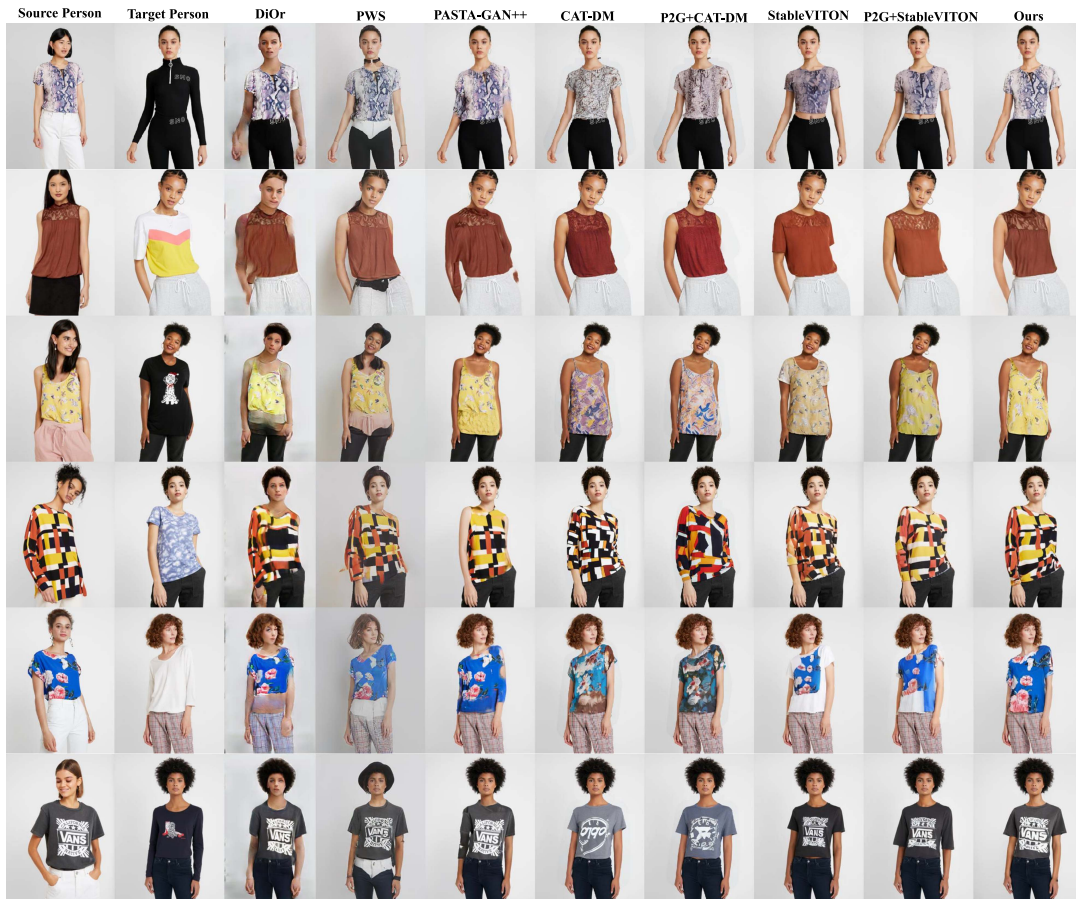


Fig. 8. Qualitative comparison for P2P try-on on VITON-HD dataset.

virtual try-on. However, Figs. 8 and 10 demonstrate that these resource-intensive pre-trained models have not yet achieved satisfactory performance levels for P2P task. Due to the absence of a dedicated module for aligning garments from the source person, which may have different sizes and positions, to the target person, directly employing StableVITON and CAT-DM

on the P2P task would result in synthesized color distortions and alterations in clothing shape. By introducing our designed plug-and-play G2P strategy, we can align garments from various source poses to an *A-Pose* that closely approximates the in-shop garment. This significantly improves the performance of resource-intensive pre-trained diffusion models on P2P tasks,

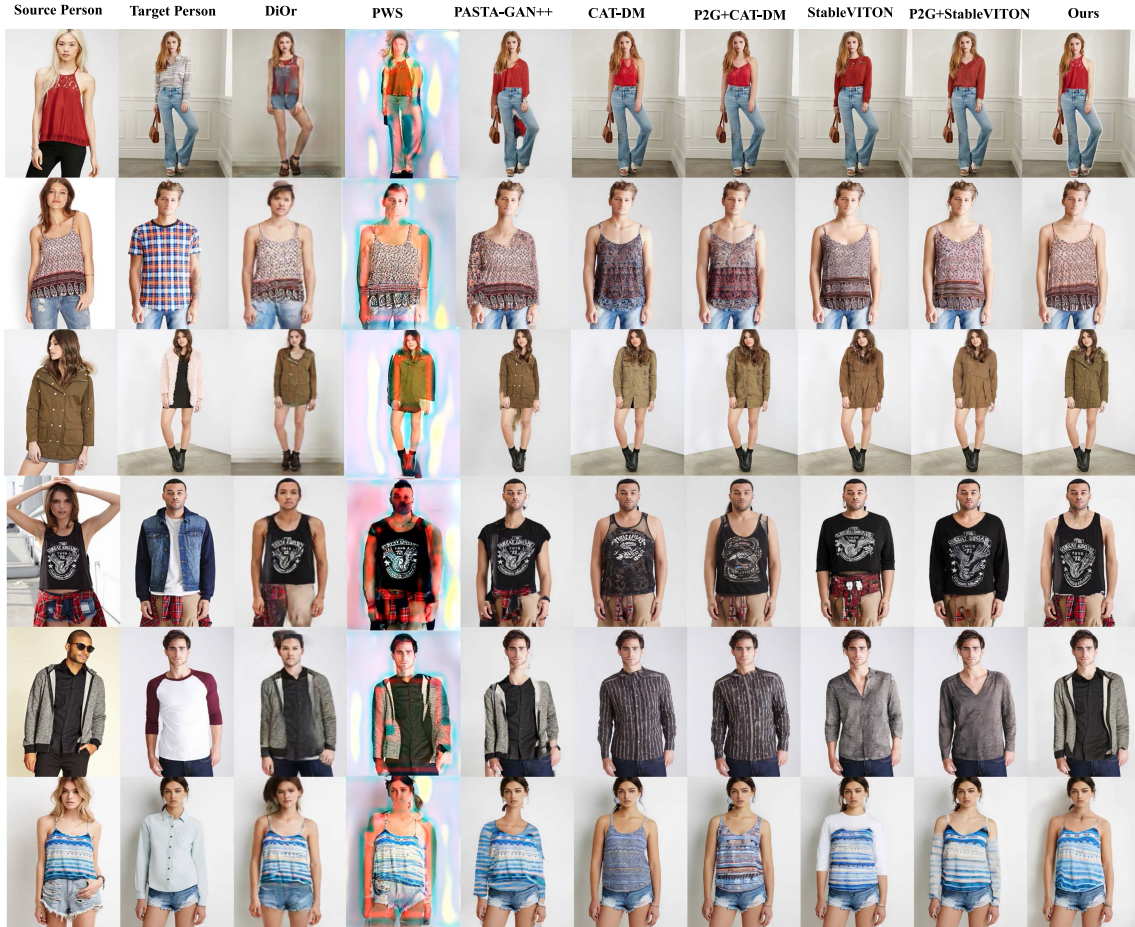


Fig. 9. Qualitative comparison for P2P try-on on Deepfashion dataset.

especially StableVITON. Furthermore, Fig. 10 clearly demonstrates that our method can accurately preserve the shape details of the dress, including the curvature of the skirt, the flowing sensation, the ruffle design, and details such as a one-shoulder design. This is something that none of the comparative methods are able to achieve. Furthermore, Fig. 9 presents the qualitative experimental results of our method on the Deepfashion dataset. It is important to note that we did not conduct unwrapping model training on the Deepfashion dataset.

G. Ablation Study

A-Pose Estimator: To evaluate the effectiveness of the *A-Pose* estimator in P2G step, we conducted training for our unwrapping module under two different *A-Pose* conditions. The first condition, referred to as “w Estimator”, involved using the *A-Pose* estimator to calculate individual *A-Pose* keypoints for each target garment. In contrast, the second condition, referred to as “w/o Estimator”, utilized a fixed front-view *A-Pose* keypoint for all target garments. The results from Fig. 11 demonstrate that the training condition using a fixed front-view *A-Pose* keypoint struggles to disentangle the garment from the source pose, whereas the condition using the *A-Pose* keypoint estimator was able to effectively remove the influence of the source pose on the garment shape. With the guidance of the accurately calculated

TABLE IV
ABLATION STUDIES OF *A-Pose* ESTIMATOR AND CLIP/DINO PATH ON VITON-HD DATASET

Method	VITON-HD	
	FID ↓	KID ↓
w/o Estimator	10.38	0.201
w/o CLIP/DINO2 Path	10.26	0.191
ViTon-GUN (CLIP)	10.06	0.167
ViTon-GUN (DINO)	10.00	0.154

A-Pose keypoints by the estimator, the garment from the source pose can be precisely aligned to a canonical representation *A-Pose*. As shown in Fig. 11, the unwrapped garment successfully preserves the details such as straps, leading to a quantitative performance improvement in P2P results, as demonstrated in Table IV.

CLIP/DINO Path: To validate the effectiveness of the CLIP/DINO path, we conducted an ablation experiment on the generator by excluding the use of the CLIP/DINO path and instead setting the input of mapping network to random noise. In Fig. 12, we visualize the generated images with and without the CLIP/DINO path. The results from the P2P comparison show that when the CLIP/DINO path is applied, the detailed features such as logos, patterns, and textures of the clothing are better preserved. This visual enhancement is also supported by the quantitative performance shown in Table IV.



Fig. 10. Qualitative comparison for P2P try-on on dresses.

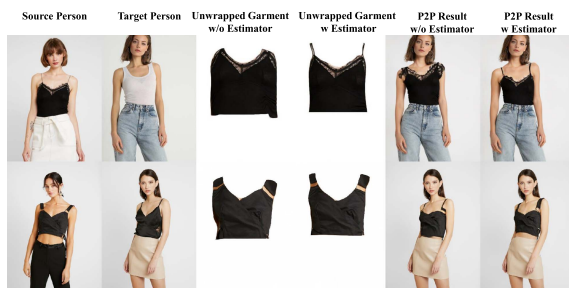


Fig. 11. Visual comparisons under the training condition with and without the *A-Pose* estimator.

V. CONCLUSION

In this work, we present a novel perspective for P2P try-on by introducing P2G scheme, which unwraps the target garment to fit an *A-Pose* while preserving its fine details. By designing the *A-Pose* estimator, we can predict *A-Pose* keypoints for each target garment, thus enabling the establishment of the necessary training conditions for the flow-based unwrapping module. Based on the P2G strategy, we propose a simpler, more straightforward, and effective pipeline for P2P virtual



Fig. 12. Visual comparisons with regard to generator utilizing the CLIP/DINO Path and otherwise.

try-on. Additionally, we design a dual-path shape-texture embedding generator to better preserve clothing details. Quantitative and qualitative experiments demonstrate that our pipeline

has achieved significant improvements, especially for dress items. Our approach successfully preserves the original shape of dresses, including the curvature of the skirt, the flowing sensation, and details like ruffled or jagged hemlines. Furthermore, our P2G strategy can serve as a plug-and-play adapter, allowing the direct application of numerous state-of-the-art pre-trained diffusion-based models trained on G2P datasets to P2P tasks. This enables achieving superior performance without the need for additional training.

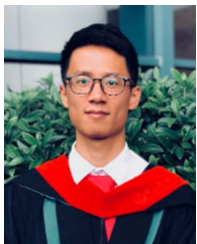
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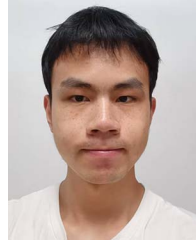
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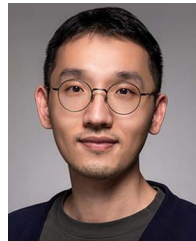
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Analysis and Machine Intelligence, etc.

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